ECONOMICS RESEARCH PAPER

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Can India's Privatisation-Led Space Policy Reforms Realistically Enable the Private Sector to Capture 8% of the Global Space Economy by 2033?

A Critical Assessment of the Gap Between Policy Promise and Commercial Reality

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Subject: Principles of Economics | Topic: The Space Economy in India

EXECUTIVE SUMMARY

India is on the cusp of a significant structural transformation in its space economy thanks to the liberalisation of the sector in 2020, the establishment of IN-SPACe, and the Indian Space Policy 2023. The government and IN-SPACe have set a bold objective of scaling India's space economy from an estimated $8.4 billion in 2022 to $44 billion by 2033, or about 8% of the projected global space economy (IN-SPACe, 2023; FICCI & EY, 2025). This paper explores whether the present privatisation-led policy approach is adequate to achieve this target.

Based on economic theory, policy analysis, industry reports, and information on private investment and development, this paper demonstrates that reforms have ignited a private space economy, but the institutional and commercial frameworks needed to attain lasting scale have not been established.

A major impediment of four structural constraints is regulatory uncertainty, limited access to risk-tolerant funding, reliance on public infrastructure, import of foreign inputs, and increased global competition. For these reasons, the hypothesis, to some extent, will not be achieved or will be achieved in conditional terms. Even though the space economy of India will expand, achieving the target of $44 billion by 2033 seems unlikely within the present system unless a new reform cycle improves legal and statutory regulation, increases long-term capital availability, supports transfer of technology, builds out domestic supply chains, and focuses on applications with commercial scale in downstream areas. In conclusion, the first cycle of reform successfully threw open the doors to the market, but this should be followed by enhancing capacity to compete on a global level through structural institutional changes.

I. ISSUE

India is at the brink of a historic transition within its space economy. From being exclusively monopolized by the Indian Space Research Organisation (ISRO)-the governmental agency globally recognised for its cost-efficient missions like Mars Orbiter Mission and Chandrayaan-3-for more than five decades, June 2020, the Indian administration embarked on comprehensive structural reforms introducing the private sector participation in Indian space economy, with culminating in 2023 into the Indian Space Policy 2023 and establishment ofIN-SPACe (Indian National Space Promotion and Authorisation Centre) as nodal regulating body. (Government of India, 2023; IN-SPACe, 2024)

The government has envisaged an ambitious target of catapulting Indian space economy to USD 44 billion by 2033 from USD 8.4 billion in 2022 and capture 8% of the global market. This will entail an average annualized growth rate of more than 15% with private sector-borne propulsion. (FICCI & EY, 2025)

However, the problem isn't with the fact, whether India has a space economy, nor with the expansion of the same. The question is, whether the framework of reform led by privatisation, structured the way it has and applied the way it has, has adequate structural underpinnings to meet the target of capturing 8% of the world market by 2033. The space economy is a complex and capital intensive sector requiring high level of technological know-how and knowledge as well as subjected to stringent regulatory and geopolitical compulsions and thus, whether the policy intention can be complemented by institutional preparedness, finance structure, regulatory clarity and industrial infrastructure-is what this paper delves into. (FICCI & EY, 2025; Government of India, 2023)

II. TOPIC

The precise topic of this research paper is:

"From Policy to Profit: Evaluating Whether India's Privatisation Framework Can Close the Gap Between Space Economy Ambition and Commercial Reality"

It has been intentionally restricted. It is not a question of whether or not the space ambition should be encouraged; India’s success here cannot be disputed. It is also not about whether India has space capabilities.

It clearly does.

It’s a more granular and perhaps more difficult economic question: Given the ground-level realities and institutional framework, will the reform framework enable India to bridge the gap between its current position (approx 2% of global space economy) and target status (8% by 2033) in the set period?

This particular framing will allow for a treatment of the dichotomous space entrepreneurship narrative of the Indian state envisioning a bright future of space for all and the Indian space startups navigating present economic realities (IN-SPACe, 2024; FICCI & EY, 2025).

III. PROBLEM STATEMENT

In the present paper, we will discuss whether the post-2020 reforms aimed at privatizing the Indian Space Sector, culminating in the Indian Space Policy 2023, the revised FDI Regime and the institutionalization of IN-SPACe are adequate enough in the current form to ensure Indian Private Space Sector achieve the government's target of achieving USD44B by 2033 or 8% of the total global market space which is to be approximately $550Bn in 2030 (Government of India, 2023; FICCI & EY, 2025) More broadly, the paper will delve into the problem from the following specific dimensions.

Firstly, we will analyze the regulatory and legal framework inadequacy of India's private space sector, particularly, the absence of an overarching dedicated ‘Indian Space Act’ and the unofficial status of IN-SPACe which lack binding legal force(Government of India, 2023; Rathore & Bhatt, 2023)

Secondly, we will examine the systemic structural funding gap of Indian start ups space-tech sector, how Indian space-tech sector is lagging behind other countries in funding trajectory and the impact of recent ad-hoc VC fund launched by the government(Tracxn, 2024; FICCI & EY, 2025).

Thirdly, we will look at the industrial infrastructure roadblock which includes India’s commercial reliance on ISRO facilities, import dependency for essential spacecraft components and the infancy of a domestic supply chain for the space sector (Invest India, 2024; Modern Diplomacy, 2024)

Fourthly, we will analyse the Competitive Dynamics of India in global commercial Launch market with existing competition such as SpaceX, Rocket Lab and new entrants like chinese launch companies(FICCI & EY, 2025; Office of Space Commerce, 2024).

The article shall neither assess ISRO’s impressive Scientific Achievements nor shall make judgments thereon; rather, it aims to focus only on the Commercial Space Economy that is considered to be directly associated with the growth of the private sector, investor funding and competition(Weinzierl, 2018).

IV. HYPOTHESIS

> **Hypothesis**: While India's space privatisation reforms of 2020–2023 have successfully catalysed the emergence of a private space ecosystem, the absence of comprehensive space legislation, the persistence of structural regulatory ambiguities, the inadequacy of domestic capital flows relative to the capital requirements of orbital-class ventures, and the dependence of the private sector on public infrastructure will collectively constrain growth below the government's USD 44 billion target by 2033, unless a second, deeper wave of institutional reform is executed with greater urgency and specificity than has been demonstrated thus far. (Government of India, 2023; FICCI & EY, 2025; Tracxn, 2024)

In simpler terms: the first wave of reforms opened the door for the private sector. But opening a door is not the same as building the road that leads to it. India's private space economy is real but underpowered relative to its ambition, and the current institutional framework alone is unlikely to be sufficient.

V. LITERATURE SURVEY

5.1 The Global New Space Economy: A Theoretical Foundation

Matthew Weinzierl's 'Space, the Final Economic Frontier' (Journal of Economic Perspectives, 2018) forms a foundational part of academic literature in comprehending the commercialisation of space. The economic argument from Weinzierl is of a "structural change" in the economics of space from "government-led to market-led models" because of reductions in launch prices, micro-miniaturisation of space technology, and rise of commercial downstream activities. He defined the core constraint to commercialisation of space as the absence of sustainable markets – which are the extent to which space generated revenues are not reliant on government procurement alone. (Weinzierl, 2018)

This can be directly relevant to India. The economic model under which ISRO traditionally operated could be considered "government-led". This transition to "market-led models" through post 2020 reform and liberalisation processes is India’s transition attempt.

As Weinzierl's framework implies, the structure for this transition may exist, but would require, over and above policy pronouncements, a development of commercially sustainable downstream markets, reliable private launch vehicle capability, and the government and private players entering into sharing arrangements of various risks. (Weinzierl, 2018; Government of India, 2023)

5.2 ISRO and the Government-Led Legacy

The main body of literature in the economics of India's space program is focused on the economic efficiency of ISRO. Narasimha (2010), in a paper published in the National Institute of Advanced Studies, highlights how efficient ISRO is economically, as it is possible for an Indian space mission to cost only a small portion of what an equivalent mission conducted by NASA or ESA costs. It was this cost efficiency that came from lower salaries, thrifty engineering techniques, and improvisation that made cost efficiency the main differentiating factor of India in the global launch services industry. (Narasimha, 2010)

However, it needs to be noted that this cost advantage came under the state-led approach, where cost efficiency was ensured because of the public-sector approach to management as opposed to market-driven competitiveness. Whether cost efficiency will continue if the private sector is involved remains a question. (Narasimha, 2010)

'Unlocking India's Space Economy: Pathways to Growth, Innovation, and Global Leadership' is the latest available report by the FICCI-EY group. The report foresees that the global space economy will grow past $1.8 trillion by 2035 and that India can make the ambitious goal of $44 billion, or 8% global share, if India continues to implement its policies and invest in the infrastructure. (FICCI & EY, 2025)

5.3 The 2020 Reforms and IN-SPACe: Institutional Literature

Much has been spoken in literature on the adequacy of India’s space governance reform. For example, Narayan and Srinivasan (2021) in their paper titled ‘Institutionalizing a Public-Private Partnership in Space Governance in India’ published by the Observer Research Foundation argued that the creation of IN-SPACe was a necessary, but insufficient measure. As per their discussion, the core of the issue was that IN-SPACe was an administrative body established within the Department of Space and has no statutory foundation, which means it does not have the right to make any binding obligations, resolve disputes or impose liability frameworks on private operators. In the field of business, regulatory credibility is fundamental and thus the non-statutory nature of IN-SPACe was recognized as one of its major shortcomings. (Narayan & Srinivasan, 2021)

The same issue was raised in the legal review of the Indian Space Policy 2023, conducted by Rathore and Bhatt (2023) and published in the Indian Journal of International Law, where the authors found inconsistencies in the

definitions used in the Indian Space Policy and lack of a proper timeline for IN-SPACe to develop the regulatory sub-frameworks outlined in the document. In their words, the Indian Space Policy 2023 is 'a welcome architecture with missing rooms'. (Rathore & Bhatt, 2023)

Similarly, in the article, the Indian Space Policy 2023 published on ScienceDirect, it is stated that 'the developments are not systematic' and 'IN-SPACe has been clothed with authority without corresponding legislative power.' (Rathore & Bhatt, 2023)

5.4 The Private Sector Ecosystem: Startups, Funding, and Milestones

The empirical literature available on India's emerging private space industry presents an image of tangible yet shaky growth. Research conducted in May 2025 by Sunil Mani, Trivandrum and Ahmedabad University showed that the total number of Indian private space companies rose from 11 in 2019 to over 400 in 2024, of which 229 are startups. The sector currently accounts for some 22,000 employees in ISRO, public-private ventures and startups. (Mani et al., 2025)

Milestones achieved by key companies in India's emerging space sector include: Skyroot Aerospace's launching of its first privately-developed rocket Vikram-S in November 2022; Agnikul Cosmos' successful test of its Agnibaan SOrTeD in May 2024 launched from India's first private launchpad at Sriharikota; and Pixxel's developing of commercial hyperspectral Earth observation satellites, including reportedly getting a USD 476 million NASA contract. (Mani et al., 2025; FICCI & EY, 2025)

However, the funding literature presents a less encouraging image. According to Tracxn's Space Tech Geo Report (2024), the total amount of money raised by India's space-tech sector amounted to USD 126 million in 2023, a record, but fell by 55% to USD 59.1 million in 2024. It coincided with the period when space investment globally began slowing down. India's government's reaction to it by approving Rs 1,000 crore (USD 114 million) VC fund under IN-SPACe in October 2024 was crucial and possibly even reactive. In contrast, Europe's space tech sector managed to raise USD 407 million in 2024. The US commercial space sector raises billions every year. (Tracxn, 2024)

The 2022 IndiaSpend analysis of funding in space startups said that out of USD 245.35 million raised by Indian space startups in seven years prior to 2022, USD 198.22 million was raised since 2020, confirming that the reforms unlocked capital for the sector. Nevertheless, the amounts raised by India's space sector in absolute terms are small compared to what is required for orbital-class launch vehicles and satellite constellations. Development of orbital-class rockets costs hundreds of millions of dollars; the total annual funding of India's space sector in its most prosperous years hardly equals one iteration of Falcon 9 development. (Business Standard, 2022))

5.5 Industrial Infrastructure and Technology Dependency

A 2024 analysis published in Modern Diplomacy by space policy researchers identified a critical and underappreciated constraint on India's private space ambitions: the absence of a domestic supply chain for key spacecraft components. The study documented that Indian space startups are dependent on imported solar panels, advanced composites, propulsion components, and certain electronics, all subject to high import tariffs that erode the cost competitiveness that has historically been India's primary global advantage. (Modern Diplomacy, 2024)

This literature positions India's private space sector as suffering from a classic industrial development dilemma: the sector cannot scale without a domestic supply chain, but the supply chain cannot develop without a scaled market. Government procurement commitments and technology transfer from ISRO can theoretically break this cycle, but the pace of ISRO technology transfer to non-governmental entities has been criticised as slow and insufficiently structured. (Modern Diplomacy, 2024; Government of India, 2023)

Invest India's 2024 sectoral report acknowledged that between 2015 and 2024, India launched 393 foreign commercial satellites, generating approximately USD 439 million in foreign exchange, an impressive figure in absolute terms, but modest relative to total global launch revenue in the same period. The report attributed India's

underperformance in capturing launch market share partially to launchpad constraints and the limited commercial launch cadence achievable through ISRO's existing infrastructure. (Invest India, 2024)

5.6 Competitive Global Context

The global competitive literature is essential context. The Morgan Stanley and Goldman Sachs projections referenced in multiple secondary sources estimate the global space economy at USD 1 trillion or more by 2040. Within this, the commercial satellite services segment is the largest and fastest growing, driven by low-Earth orbit (LEO) broadband constellations (Starlink, OneWeb, Amazon Kuiper) and Earth observation services. (World Economic Forum, 2025; Policy Circle, 2024)

SpaceX's Starlink alone has launched over 6,000 satellites and generated revenues that by some analyst estimates exceeded USD 6 billion annually by 2024, an amount nearly equal to India's entire current space economy. This contextualises the scale of ambition India is pursuing. For India to capture 8% of a USD 550 billion-plus global market, its entire space economy needs to be larger than the entire current global space economy. (World Economic Forum, 2025)

The Office of Space Commerce (US), in a 2024 consultation document, flagged specific concerns about market access challenges for American companies in India, including issues with intellectual property protection, frequency allocation processes, export control complexities, and the mandatory 50% local content requirement in Indian government space contracts. These barriers, while designed to promote domestic industry, also signal to foreign companies the friction they will face in India, friction that could slow the international partnerships India needs to accelerate its own ecosystem. (Office of Space Commerce, 2024)

VI. ANALYSIS

6.1 Applying the Literature: The First-Wave Reform Dividend

The literature collectively establishes that the post-2020 reforms generated a genuine and measurable first-wave dividend. The number of private space companies grew forty-fold in five years. Foreign and domestic investment exceeded USD 245 million. Historic milestones in private rocketry and satellite development were achieved. ISRO's infrastructure was made accessible to private players through IN-SPACe, and a legal framework was articulated through the Space Policy 2023. (FICCI & EY, 2025; Mani et al., 2025; Tracxn, 2024)

These outcomes are not trivial. Prior to 2020, India's private space sector was essentially non-existent in any commercially meaningful sense. The reforms created the conditions for a startup ecosystem to emerge, and it has emerged. On this dimension, the hypothesis would need to acknowledge that the first wave of reform succeeded in its catalytic role. (FICCI & EY, 2025; Government of India, 2023)

The critical analytical question, however, is whether a catalytic first wave is sufficient to generate the 500%+ growth required to reach USD 44 billion from USD 8.4 billion in roughly a decade. Here, the literature compels a more sceptical assessment. (FICCI & EY, 2025)

6.2 The Regulatory Gap: An Unsolved Problem

The most consistent finding across the literature is the inadequacy of India's current regulatory architecture. IN-SPACe's non-statutory status is not a technical detail, it is a structural problem with direct economic consequences. In capital-intensive industries, investors need legal certainty. They need to know that if their satellite is damaged by another party's launch vehicle, there is a clear liability framework. They need to know that spectrum rights, once allocated, cannot be revoked without due process. They need to know that disputes will be adjudicated by an independent body with genuine authority. (Narayan & Srinivasan, 2021; Rathore & Bhatt, 2023)

None of these assurances currently exist in India's space sector in a legally robust form. The Indian Space Policy 2023 acknowledged these gaps but provided neither a legislative solution nor a timeline for one. The pending national Space Bill has been 'forthcoming' for several years. This is not merely an abstract concern: the Regulatory Gaps analysis by Rathore and Bhatt (2023) documented cases where private companies faced multi-ministry clearance requirements that contradicted the 'single-window' promise of IN-SPACe. When regulation is fragmented across the Department of Space, the Ministry of Defence, the Ministry of Telecommunications, and the Ministry of Commerce, the transaction costs for private actors escalate in ways that are particularly damaging for capital-constrained startups. (Government of India, 2023; Rathore & Bhatt, 2023)

Applying the Weinzierl framework to this finding: the transition from government-led to market-led space economy requires markets to be able to function, contracts to be enforceable, property rights to be defined, liability to be assignable. India's current framework does not provide this in full. The economic consequence is not merely inconvenience; it is a higher cost of capital for Indian space startups relative to competitors in markets with clearer regulatory frameworks. (Weinzierl, 2018)

6.3 The Funding Gap: Scale and Sustainability

The funding analysis reveals a structural mismatch between the capital requirements of the ambition and the capital currently flowing into the sector. The trajectory, USD 126 million in 2023, USD 59 million in 2024, is volatile and small relative to the scale needed. Orbital-class launch vehicles cost hundreds of millions of dollars to develop to commercial readiness. Satellite constellation deployment at commercial scale requires billions. India's entire sector-wide annual private investment is insufficient to fund a single major launch vehicle development programme. (Tracxn, 2024; FICCI & EY, 2025)

The government's Rs 1,000 crore (approximately USD 114 million) VC fund, while a positive intervention, is best understood as a gap-filler rather than a structural solution. USD 114 million distributed across 200+ startups cannot replicate what sustained private capital markets provide in the United States or even in Europe. Moreover, the timing, announced in October 2024, after the funding decline had already materialised, suggests a reactive rather than anticipatory posture. (Government of India, 2024; Tracxn, 2024)

Critically, the government fund is also subject to public sector decision-making timelines and processes that may not align with the speed requirements of competitive space commercialisation. SpaceX's development velocity, from concept to first flight to commercial launch manifest, has been characterised by a private capital environment that tolerates failure and rewards speed. India's public VC fund, by contrast, is likely to apply more conservative criteria and longer decision cycles. (Government of India, 2024; Weinzierl, 2018)

One important nuance from the literature: the funding gap may be partly self-correcting if early commercial milestones attract international capital. Pixxel's USD 36 million investment from Alphabet demonstrates that international investors will commit to Indian space companies with proven technology. If Skyroot achieves a successful orbital launch, perhaps the single most important near-term commercial milestone in India's private space sector, it could trigger a significant re-rating of the sector by global venture capital. In this respect, the funding gap is real but not insurmountable. (FICCI & EY, 2025)

6.4 Industrial Infrastructure: The Supply Chain Problem

The supply chain dependency identified in the literature represents the deepest structural constraint on India's space ambitions, and the one least amenable to near-term policy fixes. Building a domestic aerospace supply chain is a decade-plus undertaking that requires consistent government procurement commitments, patient private capital, skilled workforce development, and materials science capabilities that currently exist primarily in the defence sector. (Modern Diplomacy, 2024; Invest India, 2024)

The import dependency for solar panels, advanced composites, and propulsion components is particularly problematic because it undermines the cost-competitiveness that is India's primary global selling proposition. If an Indian private launch company must import 30-40% of its components at tariff-inflated prices, its cost advantage

over SpaceX or Rocket Lab narrows considerably. The 'frugal innovation' legacy of ISRO does not automatically transfer to a private sector that cannot access ISRO's in-house manufacturing capabilities. (Modern Diplomacy, 2024)

Technology transfer from ISRO to private entities, envisioned by the Space Policy 2023, has been slow. As of early 2025, IN-SPACe had received over 658 applications for various space activities, but the technology transfer pipeline remained constrained. ISRO, understandably, has concerns about intellectual property and national security that slow the transfer process. The result is that private companies must develop technological capabilities largely from scratch, an expensive and time-consuming process. (IN-SPACe, 2024; Government of India, 2023)

An application of industrial policy theory here is instructive. South Korea's development of a semiconductor industry, and more recently its space programme, relied on active state-directed industrial policy that went well beyond regulatory liberalisation, it involved direct government procurement, co-investment in manufacturing facilities, and mandatory technology licensing arrangements. India's approach has been more hands-off: reform the regulations, set up IN-SPACe, and trust market forces. The evidence from the literature suggests this may be insufficient for a sector with such high capital barriers to entry. (FICCI & EY, 2025)

6.5 The Competitive Context: A Realistic Market Share Assessment

The 8% global market share target requires India's space economy to reach approximately USD 44 billion in a projected global market of roughly USD 550 billion by 2033. To put this in perspective: the United States currently accounts for roughly 55-60% of the global space economy. Europe accounts for around 20-25%. The 8% target would make India the world's third-largest space economy, ahead of all of Europe combined if European countries are considered individually, and on par with the entire European Space Agency ecosystem. (FICCI & EY, 2025; Policy Circle, 2024)

This is an extraordinary ambition. The global competitive landscape does not stand still while India builds its capabilities. SpaceX continues to reduce launch costs at a pace that compresses the cost-advantage window for competitors. China's commercial space sector, though less visible internationally, is growing rapidly. The United Arab Emirates and Saudi Arabia are investing heavily in space ambitions backed by sovereign wealth funds of a scale India cannot match. (World Economic Forum, 2025; FICCI & EY, 2025)

The literature suggests India's realistic competitive advantage lies in three specific areas: commercial satellite services utilising Indian demand (agriculture, disaster management, navigation, broadband connectivity for underserved regions); cost-effective Earth observation data services; and launch services for small-to-medium satellites where ISRO's proven PSLV reliability and competitive pricing remain relevant. These are real markets but they are not, on their own, a path to 8% global share. (FICCI & EY, 2025; Invest India, 2024)

The downstream sector, space applications in agriculture, climate monitoring, urban planning, defence intelligence, offers the most realistic path to rapid economic value generation. As documented in the literature, space applications already account for 75% of the sector's total value in India. Growing this segment requires less capital and regulatory infrastructure than the upstream (manufacturing and launch) segment, and India's large domestic market creates an immediate customer base. (FICCI & EY, 2025)

6.6 The Hypothesis Assessed: A Conditional Verdict

Synthesising the analytical threads above, the evidence supports the hypothesis with important qualifications. The privatisation reforms have genuinely catalysed an ecosystem. But the ecosystem, as currently constituted, is not on a trajectory that credibly reaches USD 44 billion by 2033. The structural gaps, in regulation, funding, supply chain, and competitive positioning, are significant and not fully addressed by the current policy framework. (FICCI & EY, 2025; Tracxn, 2024; Government of India, 2023)

However, the hypothesis also requires a nuance the literature supports: the gap is not unbridgeable, and the trajectory is not fixed. If India passes a comprehensive Space Act with genuine statutory authority for IN-SPACe, if

a second funding wave arrives on the back of early commercial orbital success, if ISRO technology transfer is accelerated and domestic supply chains begin to develop, and if India's downstream space applications market grows as rapidly as its demographic and economic fundamentals suggest it should, then a significantly higher figure than the 2022 baseline is achievable by 2033. (Government of India, 2023; IN-SPACe, 2024; FICCI & EY, 2025)

The question is not whether India will grow its space economy. It will. The question is whether the growth will reach 8% of global share by 2033, or whether it will achieve something more modest, perhaps 4-5%, that is still historically significant but falls short of the political target. (FICCI & EY, 2025)VII. FINDINGS

Finding 1: The Reform Framework Has Successfully Created an Ecosystem, But Not a Self-Sustaining Market

The empirical evidence is unambiguous that the post-2020 reforms created a genuine private space ecosystem where none existed before. Four hundred private companies, record-breaking startup funding in 2023, historic private launches, international investor interest, these are real outcomes. The reform framework deserves credit for catalysing this. However, the literature and analysis collectively suggest that the ecosystem is still dependent on government infrastructure, government procurement, and government-facilitated access to technology. A self-sustaining commercial market, where private demand, private capital, and private infrastructure reinforce each other, has not yet emerged. (FICCI & EY, 2025; Mani et al., 2025; Tracxn, 2024)

Finding 2: The Regulatory Architecture Remains Structurally Incomplete

The absence of a national Space Act and the non-statutory status of IN-SPACe constitute a genuine structural deficit in India's space governance. This finding is consistent across the academic literature, legal analysis, and industry commentary. The economic consequences, higher cost of capital, reduced investor certainty, multi-ministry regulatory friction, are real and measurable. Unless India enacts comprehensive space legislation in the near term, this deficit will compound as the sector grows and commercial disputes become more complex. (Government of India, 2023; Narayan & Srinivasan, 2021; Rathore & Bhatt, 2023)

Finding 3: The Funding Gap Is Structural, Not Cyclical

The decline in private space funding in 2024 (down 55% year-on-year) is partly attributable to the global tech funding slowdown, but the underlying structural issue is deeper: India's domestic capital markets are not yet equipped to fund the development of orbital-class launch systems. The government's VC fund is a necessary but insufficient response. The funding gap will only close if one of three things happens: a major commercial success attracts a new wave of international capital; domestic institutional investors (insurance funds, pension funds, sovereign wealth) are encouraged or mandated to allocate a portion of assets to deep-technology sectors including space; or the government provides direct co-investment at a scale substantially larger than the Rs 1,000 crore fund. None of these is currently assured. (Tracxn, 2024; Government of India, 2024)

Finding 4: The 8% Target Is Aspirational Rather Than Analytical

The finding from the literature is that the USD 44 billion / 8% target was set on the basis of macroeconomic trajectory modelling and policy ambition rather than a granular bottoms-up analysis of the industrial, regulatory, and commercial conditions required to achieve it. This does not make the target worthless, targets serve important signalling and coordination functions, but it means the target should be understood as a directional aspiration, not a contractual commitment backed by a credible delivery roadmap. The gap between aspiration and roadmap is itself an economic problem, as it may lead to resource misallocation if policy instruments are calibrated to the aspirational target rather than to realistic interim milestones. (FICCI & EY, 2025)

Finding 5: The Downstream Space Economy Offers the Strongest Near-Term Growth Path

The finding most supported by economic logic and empirical evidence is that India's fastest route to a large space economy runs through downstream applications rather than upstream manufacturing and launch. Agriculture, disaster management, navigation, climate monitoring, and urban management applications of space-derived data are already generating substantial value, already account for 75% of the sector's value, and are addressable with less capital and regulatory complexity than orbital-class launch vehicles. A policy strategy that double-downs on the downstream while building the upstream more patiently may be more economically rational than the current framework, which treats the sector as a uniform whole. (FICCI & EY, 2025)

Finding 6: The Global Competitive Context Makes the 8% Target Harder, Not Easier, Over Time

The literature on global space competition reveals that India's window of cost-advantage is narrowing. SpaceX's continued reduction of launch costs, the emergence of Chinese commercial launch capabilities, and the massive capital flows into satellite mega-constellations from Starlink and Amazon Kuiper are reshaping the market at a pace that may disadvantage late entrants. India's competitive advantage, frugal engineering and cost efficiency, is real but potentially transient. The policy implication is that speed of reform implementation matters: delays in the Space Act, in technology transfer, and in commercial launch cadence all carry an opportunity cost that compounds over time. (World Economic Forum, 2025; FICCI & EY, 2025)

VIII. CONCLUSION

This paper set out to examine whether India's privatisation-led space economy reform framework is adequate to deliver the government's target of USD 44 billion in space economy value, representing 8% of the global market, by 2033. The hypothesis was that while the reforms have successfully catalysed an ecosystem, structural gaps in regulation, funding, industrial infrastructure, and competitive positioning will constrain growth below the target unless a deeper second wave of reform is executed urgently. (FICCI & EY, 2025)

The literature survey, drawing on economic theory, legal analysis, institutional assessments, and empirical data from the sector, strongly supports this hypothesis. The evidence points to a real but underpowered private space sector, operating within a regulatory framework that is progressively structured but legislatively incomplete, funded at levels substantially below those required for orbital-class commercialisation, and dependent on industrial infrastructure that does not yet exist in domestic form. (Weinzierl, 2018; Government of India, 2023; Tracxn, 2024; Invest India, 2024)

At the same time, the analysis resists a pessimistic conclusion. India's trajectory is genuinely impressive relative to its starting point five years ago. The demographic and economic fundamentals, a large and growing domestic market for space-based services, a deep pool of technical talent, a proven track record in cost-effective space operations, and a government with demonstrated political will to reform, create real long-term potential. The USD 44 billion target, even if not achieved precisely by 2033, is not a fantasy. It may simply be achieved in 2035 or 2037 instead. (FICCI & EY, 2025)

What the findings suggest, most urgently, is the need for a second-generation reform agenda. The first generation opened the door. The second generation must build the road. This requires: the early passage of a comprehensive national Space Act with statutory authority for IN-SPACe; a sustained and substantially larger government co-investment commitment in launch infrastructure and spacecraft manufacturing; an accelerated ISRO technology transfer programme with defined timelines and commercially structured terms; a focused industrial policy to develop a domestic space supply chain, potentially modelled on India's successful defence indigenisation programmes; and an international partnership strategy that attracts not just investment but manufacturing and technology transfer agreements from established space nations. (Government of India, 2023; IN-SPACe, 2024; FICCI & EY, 2025)

Whether India achieves 8% of the global space economy by 2033 or 4% by 2035 matters less, in the long run, than whether it builds the institutional, financial, and industrial foundations for a genuinely competitive commercial space sector. ISRO's legacy has given India a credible launchpad, in the literal and metaphorical sense. What the

private sector needs now is not just more policy ambition, but more policy architecture. The space is there. The question is whether the structures can keep up with the stars. (FICCI & EY, 2025)

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